	GLOBAL ECONOMICS **Grade: 10 - Learning Evidence 3** **DECISION-MAKING UNDER UNCERTAINTY** **Teacher: Nicolás López Cuéllar**	**1st TERM** **2026–2027**

Learning objective: Compare decision criteria and their limitations, and construct and interpret decision trees for staged filtering under uncertainty.	**Assessment criteria:** C8: Compares the Maximax, Maximin, Minimax Regret, and EMV/Bayes decision criteria by contrasting their information requirements, risk orientations, recommended alternatives, and limitations. C9: Constructs and interprets decision trees for staged filtering rules, applying decision criteria in the specified order to identify the final alternative.

Name: _____ **Grade:** _____ **Date:** _____

Criteria assessment

This exam assesses C8 and C9. Answer individually on separate paper. Questions 1–2 assess C8; each question is worth 3 points: 1 point for task (a) and 2 points for task (b). Questions 3–5 assess C9; each problem is worth 2 points: 1 point for the filtering tree and interpretation and 1 point for the supporting table. Each criterion totals 6 points. A **minimum of 4 points** is required to pass each criterion. Partial credit is available for correct comparisons, filtering stages, and justified reasoning even if later conclusions contain errors. The final C10 section is an unscored reminder of the separate activity.

Question	C8	C9
1	3	–
2	3	–
3	–	2
4	–	2
5	–	2
Total	**6**	**6**

1. Repair Workshop Plans [C8-3]

A bicycle repair workshop chooses its capacity for the next quarter before demand is known. Profits are in thousand USD; forecast probabilities are shown below.

Alternative	High	Normal	Low
$P(S_j)$	0.2	0.6	0.2
Large (A_1)	120	25	-20
Flexible (A_2)	85	65	10
Basic (A_3)	50	45	35
Partner (A_4)	70	75	20

The following table summarizes the decision-making statistics for each alternative.

	Max	Min	Regret	EMV
A_1	120	-20	55	35
A_2	85	10	35	58
A_3	50	35	70	44
A_4	75	20	50	63

(a) Report each criterion's recommended alternative(s), deciding statistic, and value. **[1]**

(b) A subsidy s (integer, $0 \le s \le 30$, thousand USD) increases the low-demand payoffs of both A_1 and A_2 by s. The rest remain unchanged. An alternative is eligible only if its profit is nonnegative in every state. Determine the ranges of s for which each alternative is eligible. Find the smallest s making A_2 the unique Maximin choice. At that s, report all four recommendations and deciding values among eligible alternatives. **[2]**

2. Community Market Stall [C8-3]

An organizer chooses a stall for one weekend. Profits are in hundred USD. The forecast assigns high attendance probability 0.40 and low attendance probability 0.60.

Alternative	High	Low
$P(S_j)$	0.4	0.6
Large (C_1)	80	10
Compact (C_2)	48	32

	Max	Min	Regret	EMV
C_1	80	10	22	38
C_2	48	32	32	38.4

(a) For part (a) only, the forecast is withdrawn as unreliable. Compare the information needed by all four criteria: which summaries still support a choice, which no longer do, and why would equal probabilities need justification? **[1]**

(b) For this part, restore the supplied probabilities as reliable. A sponsor guarantees C_2 a high-attendance profit of g, replacing 48, where g is an integer, $48 \leq g \leq 80$, in hundred USD. All other payoffs stay fixed; both stalls remain eligible. Find the smallest g making C_2 the unique Minimax Regret choice. Justify the threshold and boundary tie; interpret the increase from 48 in USD. At that g, report all four updated recommendations and deciding values, including every tie. **[2]**

Staged filtering: Questions 3–5

For **each** problem, construct a compact filtering tree with branches for retained and eliminated alternatives. Label every stage with its rule and relevant numerical values. Identify the final choice, including any tie, and write one sentence interpreting the result in context. **[1]**

Also construct a supporting table with the following columns, recording every stage and its relevant values. **[1]**

# Step	Description	Left Alternatives	Eliminated Alternatives

- Step 0 includes all six alternatives.
- Each stage compares only survivors from the previous stage. Eliminated alternatives cannot reenter.
- Supplied summary statistics remain fixed throughout filtering. Do not recalculate regrets or other statistics after eliminations.
- When keeping the best k alternatives, retain all tied at the cutoff, even if more than k survive.
- Report any final tie explicitly; do not invent a tie-breaker.

Use highest EMV for EMV/Bayes, highest Max for Maximax, highest Min for Maximin, and lowest Regret for Minimax Regret. Regrets were calculated against all six original alternatives.

3. Book Fair Stock [C9-2]

A bookseller chooses a stock plan D_1–D_6 for a fair. Supplied summaries describe net profits in hundred USD under high, normal, and low sales, with probabilities 0.50, 0.30, and 0.20.

	EMV	Max	Min	Regret
D_1	65	100	0	40
D_2	65	90	10	30
D_3	65	80	20	20
D_4	59	70	30	30
D_5	53	60	40	40
D_6	43	50	30	50

Apply these stages in order:

1. EMV/Bayes: keep the 2 highest EMVs, retaining cutoff ties.
2. Maximin: select the highest Min among survivors.

Complete the tree, table, and interpretation described above.

4. Refill Shop Expansion [C9-2]

A refill shop chooses an expansion plan F_1–F_6 for the next year. Summaries show net profits in thousand USD under strong, steady, and weak demand, with probabilities 0.30, 0.50, and 0.20.

	EMV	Max	Min	Regret
F_1	61	120	0	45
F_2	64.5	105	15	30
F_3	68.5	95	25	25
F_4	70	85	35	35
F_5	64	75	45	45
F_6	55	65	40	55

Apply these stages in order:

1. Minimax Regret: keep the 5 lowest Regret values.
2. EMV/Bayes: keep the 4 highest EMVs among survivors.
3. Maximax: keep the 2 highest Max values among survivors.
4. Maximin: select the highest Min among survivors.

Construct the filtering tree and supporting table; include rules, values, eliminations, final choice, and a one-sentence interpretation. Use the shared rules for Questions 3–5.

5. School Supply Orders [C9-2]

A school supplier chooses an order plan H_1–H_6 for the next term. Summaries show net profits in hundred USD under high, normal, and low demand, with probabilities 0.30, 0.50, and 0.20.

	EMV	Max	Min	Regret
H_1	61	120	0	45
H_2	64.5	105	15	30
H_3	68.5	95	25	25
H_4	70	85	35	35
H_5	64	75	45	45
H_6	55	65	40	55

Apply these stages in order:

1. Maximin: keep the 4 highest Min values.
2. Minimax Regret: keep the 3 lowest Regret values among survivors.
3. Maximax: keep the 2 highest Max values among survivors.
4. EMV/Bayes: select the highest EMV among survivors.

Complete the tree, table, and interpretation described above. Use the shared rules for Questions 3–5.

C10: Separate activity reminder (unscored)

C10 is assessed through the separate **Interactive Decision Webpage: C10 Practice** activity:

`book2026-2027/grade10/term_1/1-practice_activities/G10_T1_L3_C10_practice_activity.tex`

Complete the individual interactive decision webpage described in that activity and submit both the **deployed webpage URL** and the **GitHub repository URL**. Refer to that activity for the full instructions. C10 has no exam questions or points.